CareerTrack

CSCE 547 | Minor Project 4 — Web Application

Instructor: Professor Robert Bailey

Student: Divyashanu Swain Submission date: October 23rd, 2026

GitHub repository: https://github.com/dswain07/CareerTrack

1. Project Overview

CareerTrack is a job application tracking web application developed with ASP.NET Core MVC on .NET 10. It helps users organize companies, job applications, and interviews, while managing application statuses through an Administration page. The application stores data in SQLite using Entity Framework Core and supports light and dark themes that can be switched without restarting the application.

2. Technology and Architecture

Web framework	**ASP.NET Core MVC (.NET 10), Razor views**
Database and ORM	SQLite and Entity Framework Core
Data layer	CareerTrack.Data — separate .NET Class Library
Frontend	Bootstrap 5.3.3, CSS, JavaScript
Development environment	Visual Studio 2026 on Windows
Database entities	Companies, JobApplications, Interviews, Contacts, ApplicationStatuses

The database contains five application tables and 40 sample records: 10 companies, 15 job applications, 5 interviews, 3 contacts, and 7 application statuses. DDL.sql and DML.sql are included in the Scripts folder. Both scripts were tested against a fresh SQLite database, with all 40 records restored and no foreign-key violations.

3. Implemented Features

- Full Create, Read, Update, and Delete operations for Companies, Job Applications, and Interviews.
- Job Application search by Job Title and Location, including combined partial-text searching.
- Administration page for creating, editing, and deleting application statuses, with a safeguard against deleting statuses in use.
- Settings page with immediate light/dark theme switching; the browser saves the selected preference.
- Dashboard displaying company, application, interview, and offer counts, plus recent applications.

4. Required Questions

4.1 Describe a challenge you overcame in this project

One challenge was keeping the database records and MVC pages consistent while expanding the Job Applications feature. The original sample applications did not display values for Location, Job URL, Salary, and Notes. I updated the seeded records and verified that the information appeared correctly in the application. I also needed to preserve the relationships among companies, application statuses, job applications, and interviews, particularly when records were deleted. This helped me understand the importance of data integrity and testing changes across multiple layers of an application.

4.2 Describe a technique you learned in this project

I learned how to use Entity Framework Core with a separate Class Library data layer in an ASP.NET Core MVC solution. I used a DbContext, entity models, migrations, and dependency injection to connect controllers to SQLite. I also learned to implement asynchronous database queries and two-field searching with LINQ, and to generate and verify DDL and DML scripts for database reconstruction.

4.3 What would you refine to make the project ready to sell, ship, and deploy?

Before production deployment, I would add user authentication and role-based authorization so that each user can access only their own applications and only authorized administrators can change statuses. I would also improve input validation, error handling, automated testing, accessibility, and mobile responsiveness. Additional useful features would include reminders for follow-ups and interviews, filtering and sorting, and secure deployment with monitoring and backups. The current Administration page is a demonstration feature and does not enforce access control.

4.4 Which technology did you choose and why?

I chose ASP.NET Core MVC because it provides a clear separation between models, views, and controllers. This structure makes it straightforward to organize database logic, request handling, and the user interface. MVC also works well with Entity Framework Core, Razor views, and Bootstrap, allowing me to build a structured, maintainable web application using C# and .NET 10.

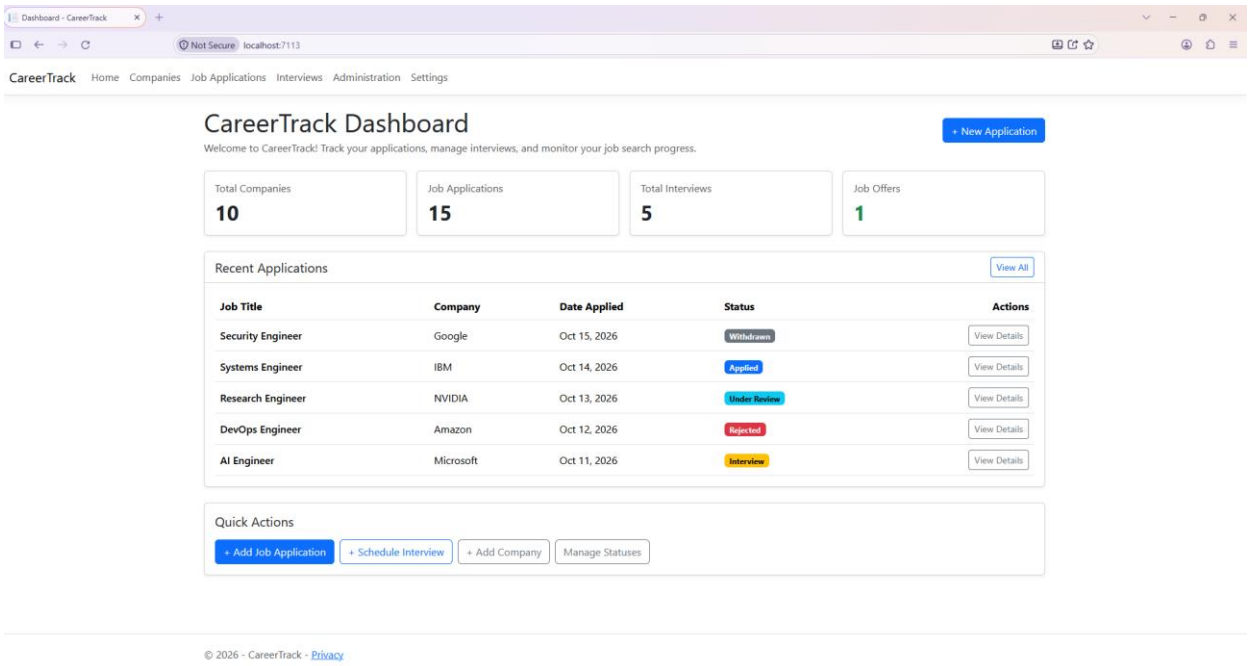
5. Graduate Requirement: Separate Data Layer

I created CareerTrack.Data as a separate Class Library project within the CareerTrack solution. I placed the entity models, CareerTrackDbContext, migrations, and database seeding logic in that library. I added a project reference from the CareerTrack MVC web project to CareerTrack.Data, configured the SQLite connection string in the web project, and registered CareerTrackDbContext with dependency injection in Program.cs. Controllers can then receive the DbContext through their constructors. EF Core migrations use the data-layer project together with the web application as the startup project. This arrangement keeps persistence code separate from the presentation layer.

6. Screenshot Evidence

Replace each marked area with an actual screenshot captured from the running CareerTrack application. Screenshots should show the browser UI and relevant action or result. The first four are explicitly required by the assignment.

1 — Homepage



2 — Search

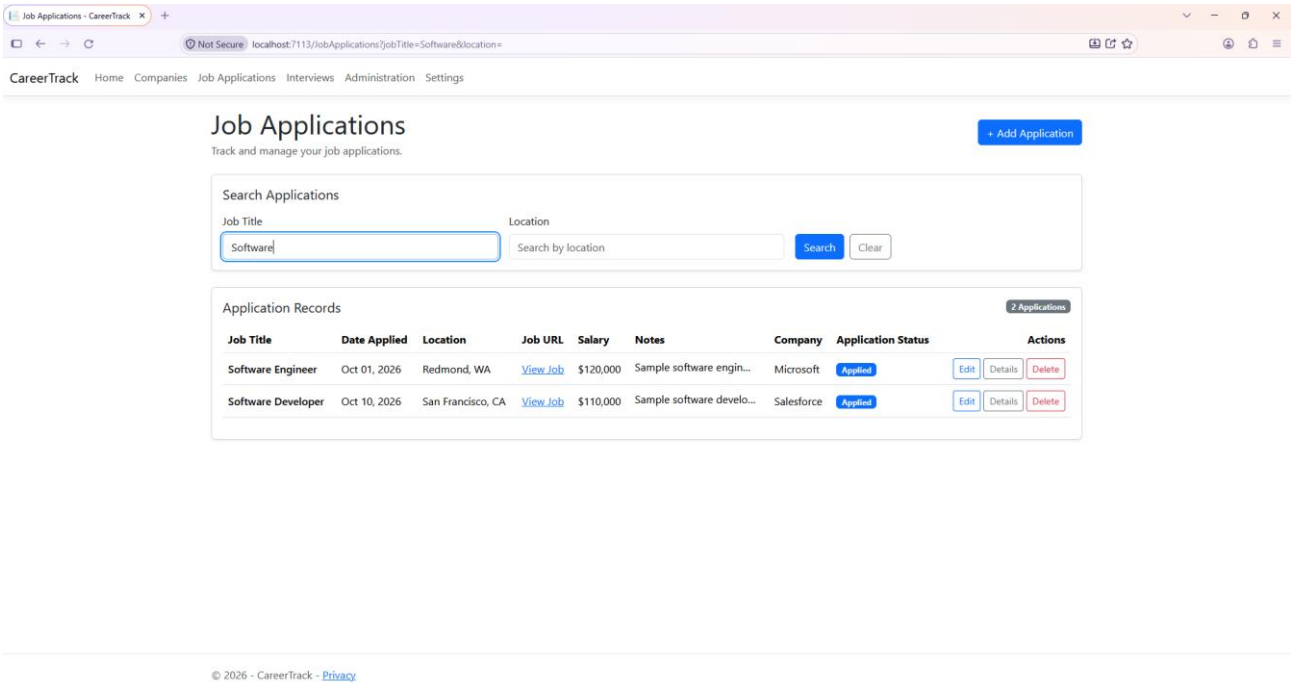


Figure 1. Search by Job Title

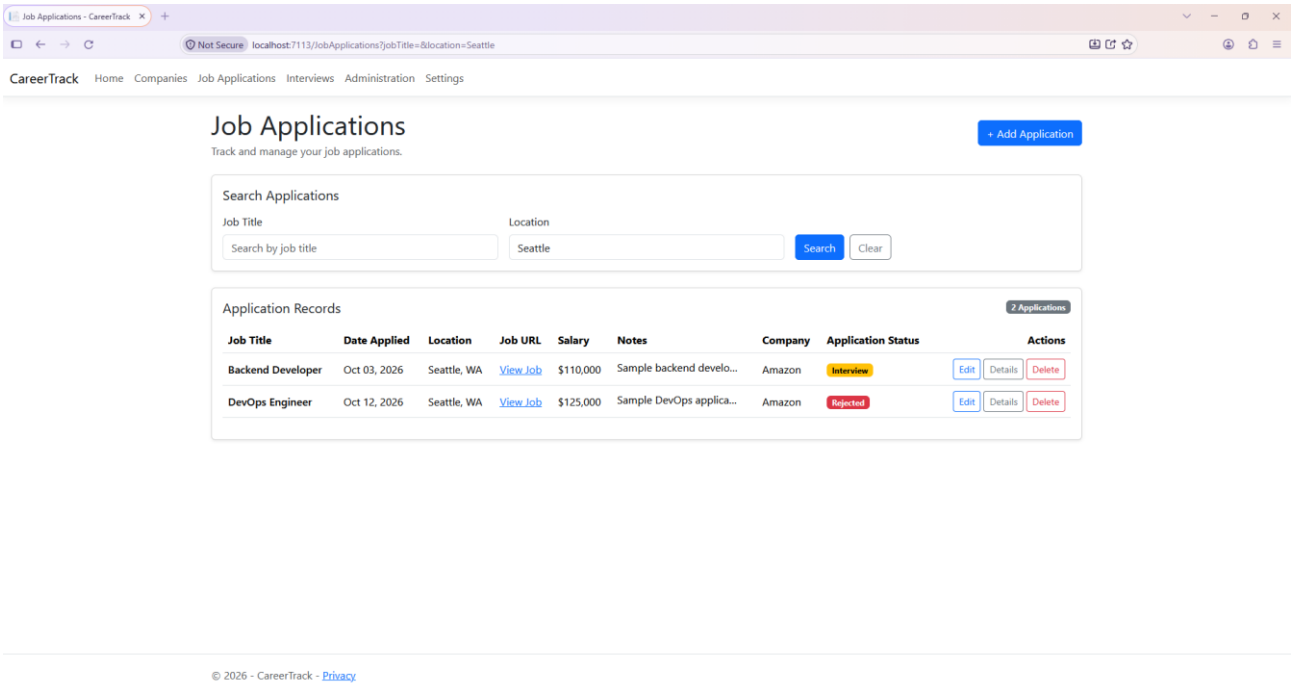


Figure 2. Search by Location

3 — CRUD on Companies Table

3.1 Create

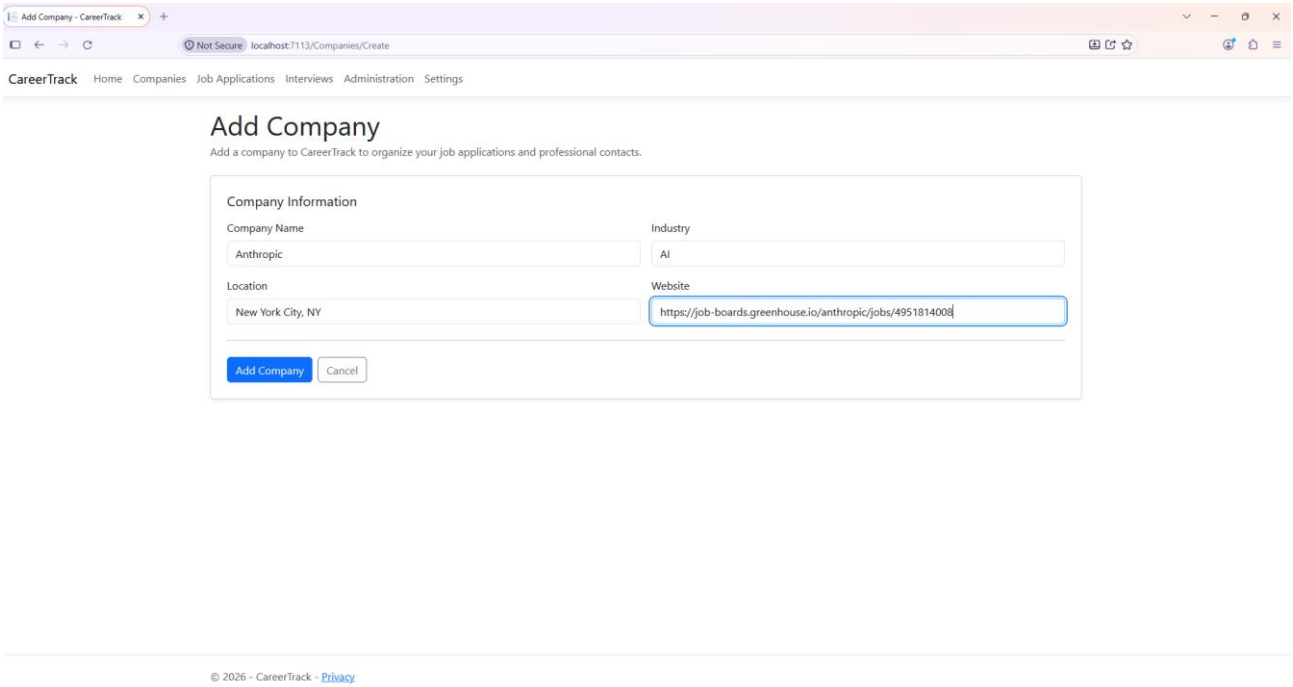


Figure 3. Create Company

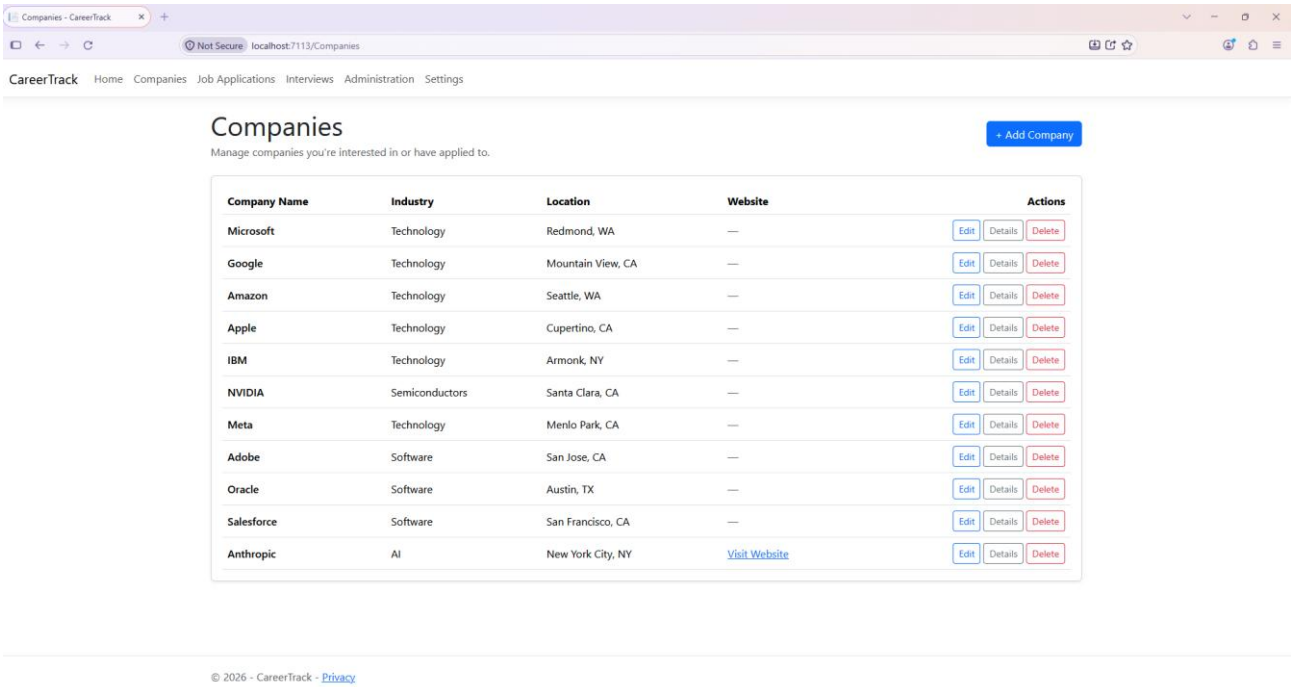


Figure 4. Company Created

3.2 Update

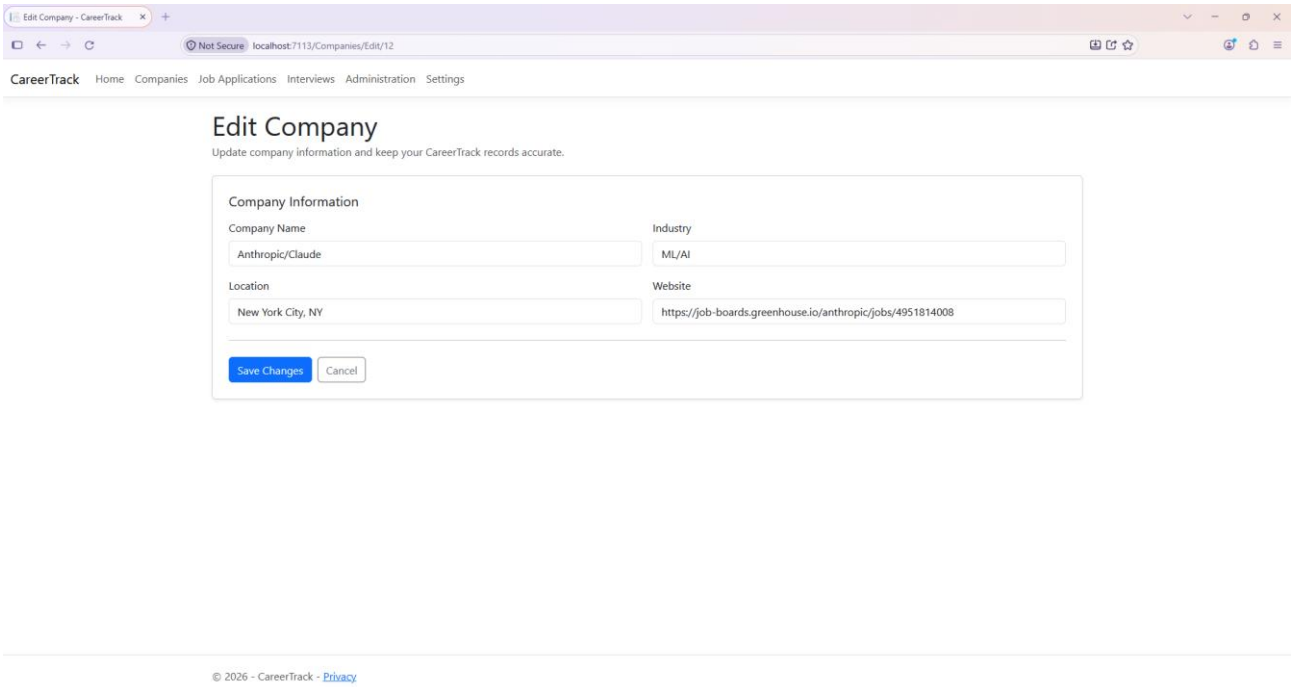


Figure 5. Update Company

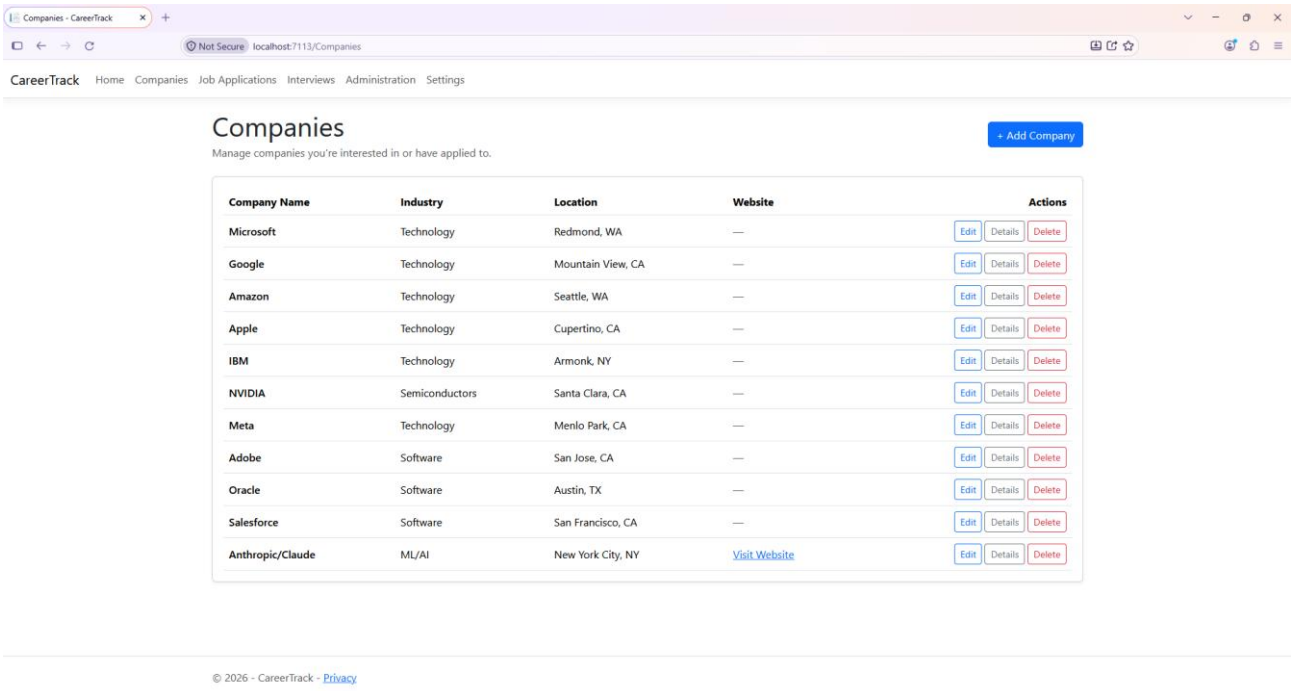


Figure 6. Company Updated

3.3 Read

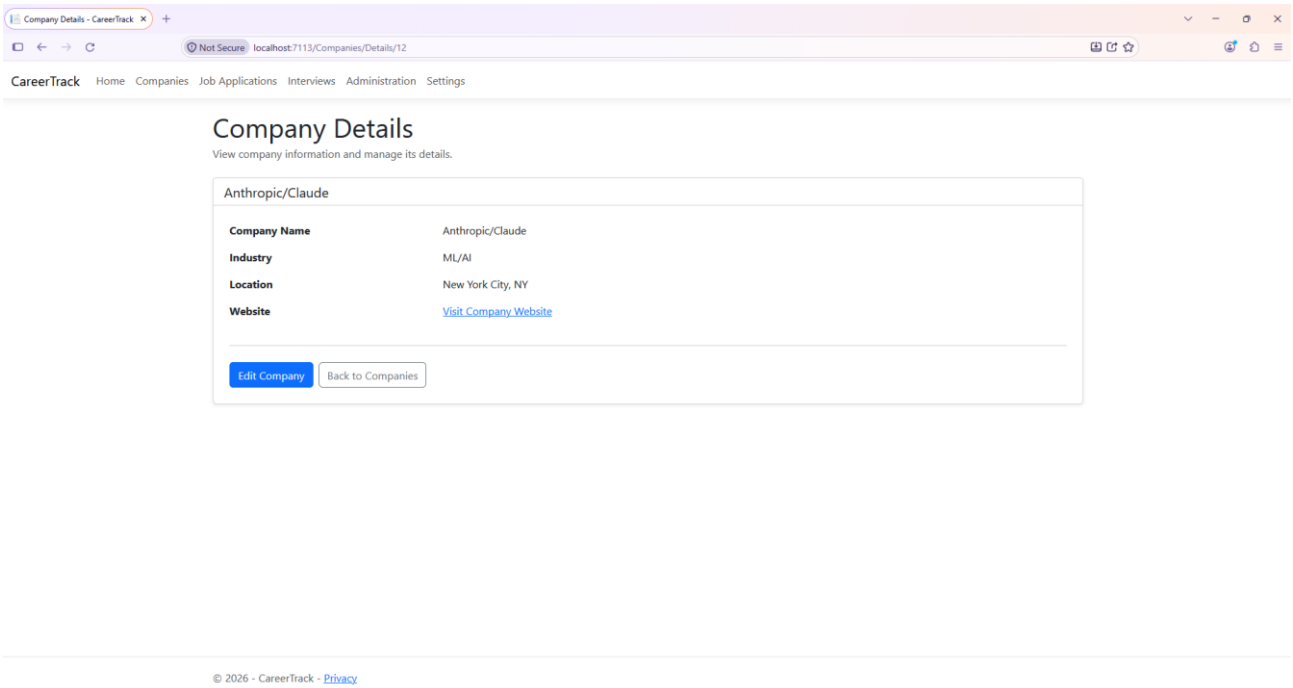


Figure 7. Read Company

3.4 Delete

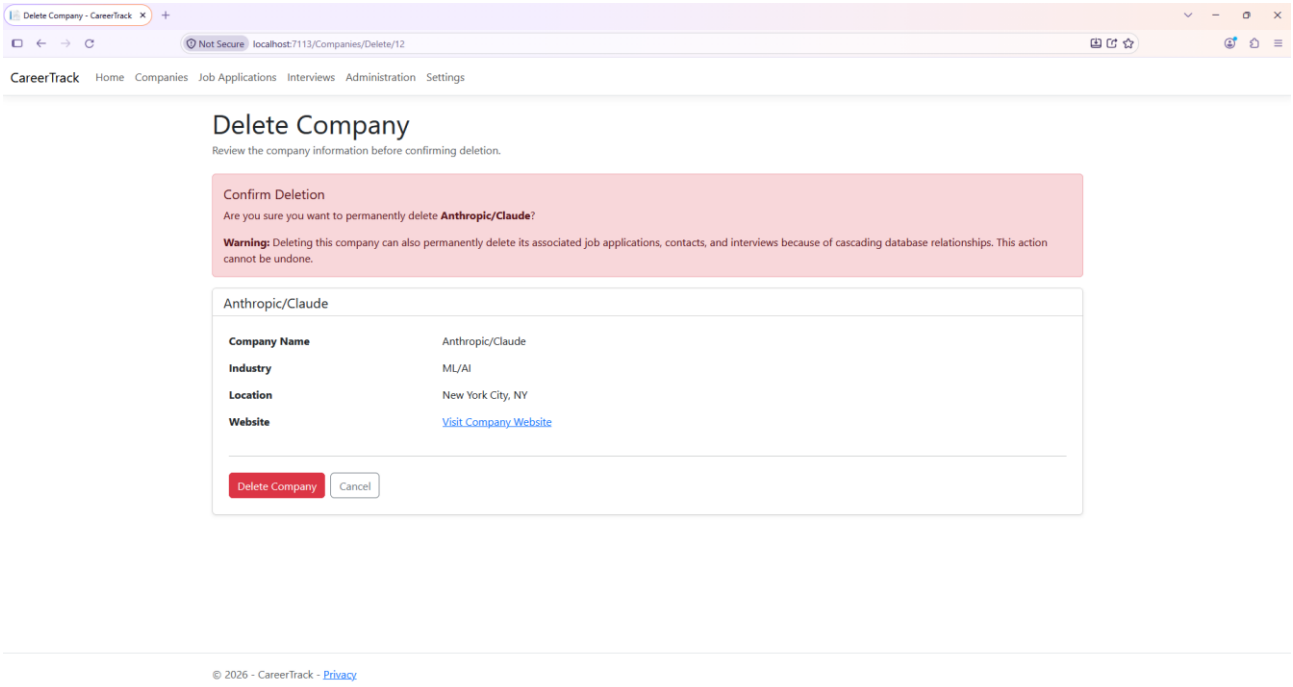


Figure 8. Delete Company

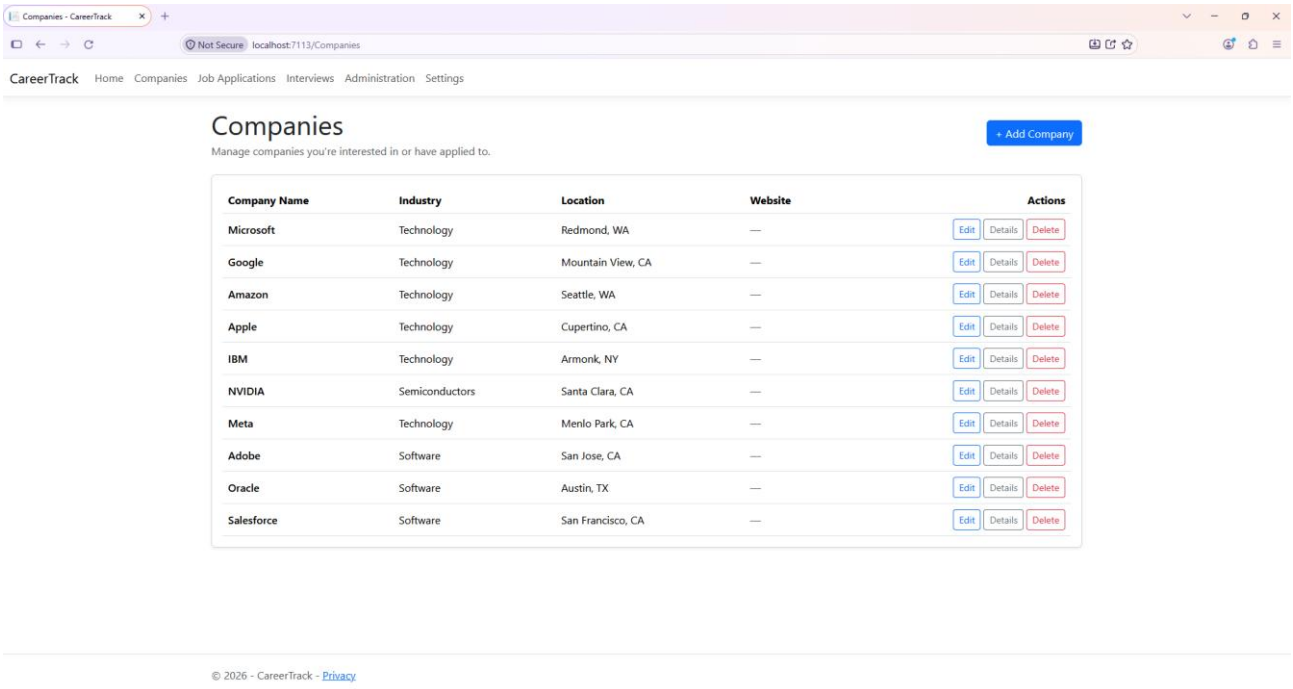


Figure 9. Company Deleted

4 — CRUD on Job Application Table

4.1 Create

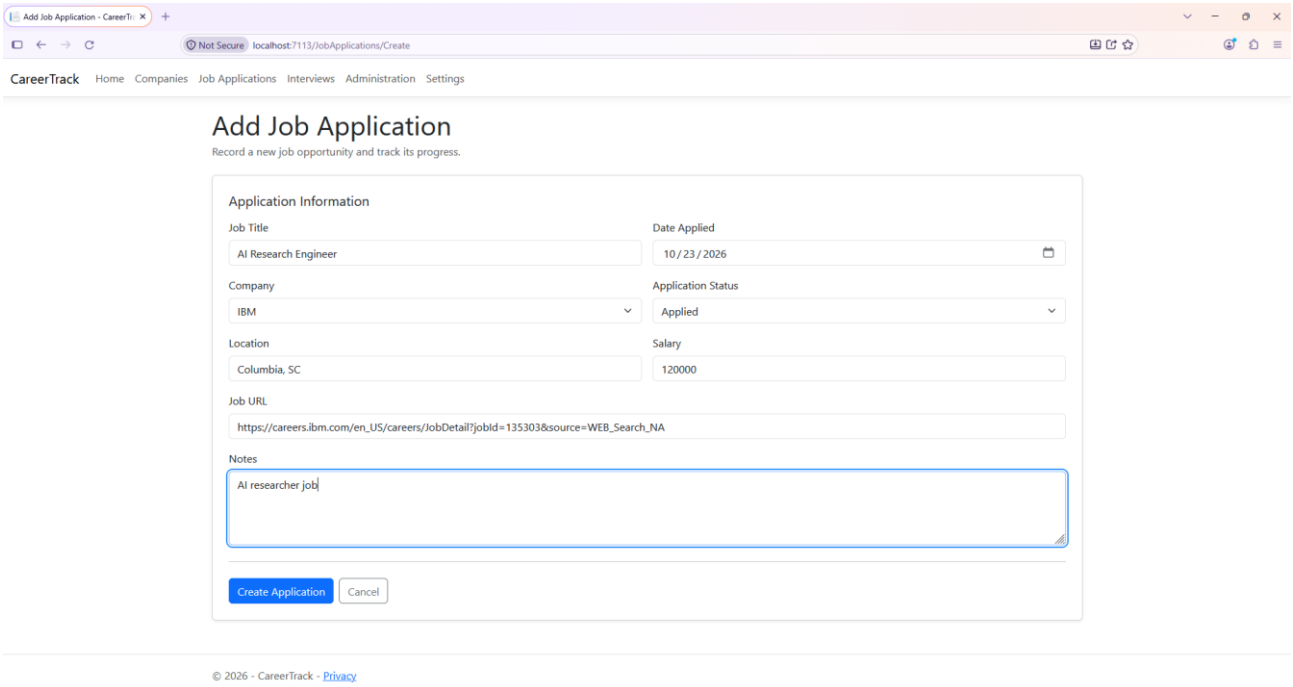
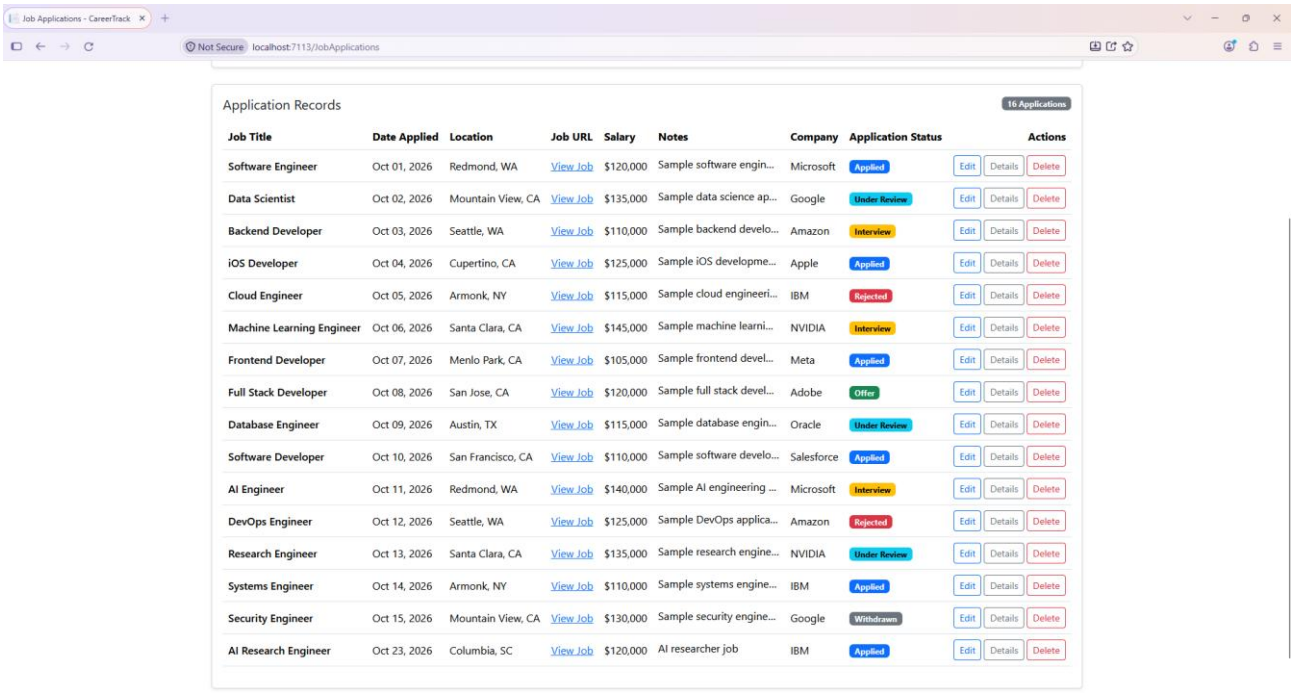


Figure 10. Create Job Application



Job Title	Date Applied	Location	Job URL	Salary	Notes	Company	Application Status	Actions
Software Engineer	Oct 01, 2026	Redmond, WA	View Job	$120,000	Sample software engin...	Microsoft	Applied	Edit Details Delete
Data Scientist	Oct 02, 2026	Mountain View, CA	View Job	$135,000	Sample data science ap...	Google	Under Review	Edit Details Delete
Backend Developer	Oct 03, 2026	Seattle, WA	View Job	$110,000	Sample backend develo...	Amazon	Interview	Edit Details Delete
iOS Developer	Oct 04, 2026	Cupertino, CA	View Job	$125,000	Sample iOS developme...	Apple	Applied	Edit Details Delete
Cloud Engineer	Oct 05, 2026	Armonk, NY	View Job	$115,000	Sample cloud engineeri...	IBM	Rejected	Edit Details Delete
Machine Learning Engineer	Oct 06, 2026	Santa Clara, CA	View Job	$145,000	Sample machine learni...	NVIDIA	Interview	Edit Details Delete
Frontend Developer	Oct 07, 2026	Menlo Park, CA	View Job	$105,000	Sample frontend devel...	Meta	Applied	Edit Details Delete
Full Stack Developer	Oct 08, 2026	San Jose, CA	View Job	$120,000	Sample full stack devel...	Adobe	Offer	Edit Details Delete
Database Engineer	Oct 09, 2026	Austin, TX	View Job	$115,000	Sample database engin...	Oracle	Under Review	Edit Details Delete
Software Developer	Oct 10, 2026	San Francisco, CA	View Job	$110,000	Sample software develo...	Salesforce	Applied	Edit Details Delete
AI Engineer	Oct 11, 2026	Redmond, WA	View Job	$140,000	Sample AI engineering ...	Microsoft	Interview	Edit Details Delete
DevOps Engineer	Oct 12, 2026	Seattle, WA	View Job	$125,000	Sample DevOps applica...	Amazon	Rejected	Edit Details Delete
Research Engineer	Oct 13, 2026	Santa Clara, CA	View Job	$135,000	Sample research engine...	NVIDIA	Under Review	Edit Details Delete
Systems Engineer	Oct 14, 2026	Armonk, NY	View Job	$110,000	Sample systems engine...	IBM	Applied	Edit Details Delete
Security Engineer	Oct 15, 2026	Mountain View, CA	View Job	$130,000	Sample security engine...	Google	Withdrawn	Edit Details Delete
AI Research Engineer	Oct 23, 2026	Columbia, SC	View Job	$120,000	AI researcher job	IBM	Applied	Edit Details Delete

Figure 11. Job Application Created

4.2 Update

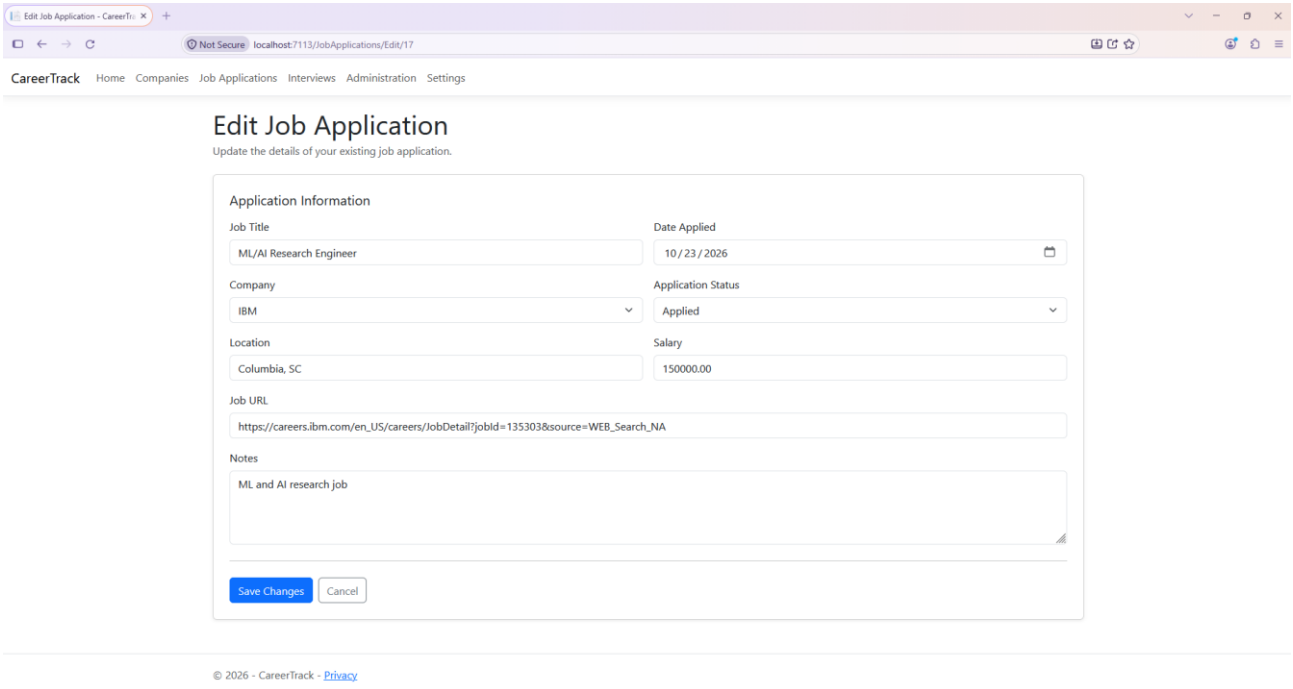


Figure 12. Update Job Application

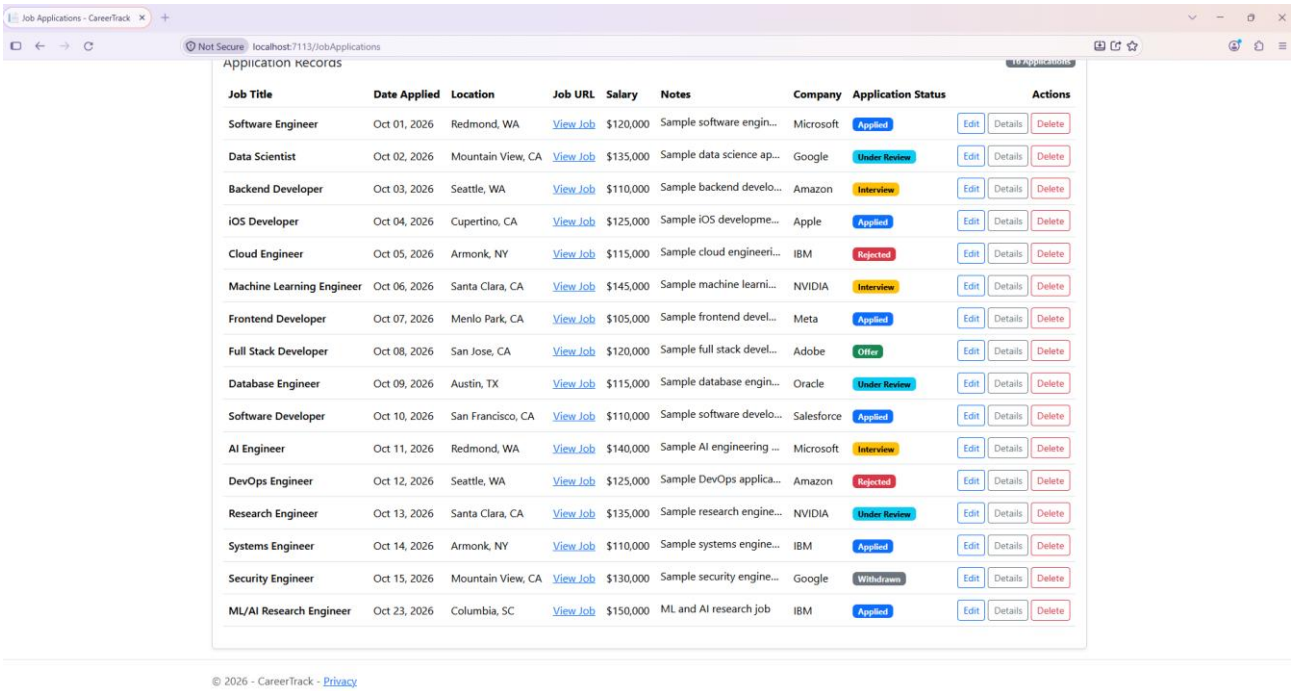


Figure 13. Job Application Created

4.3 Read

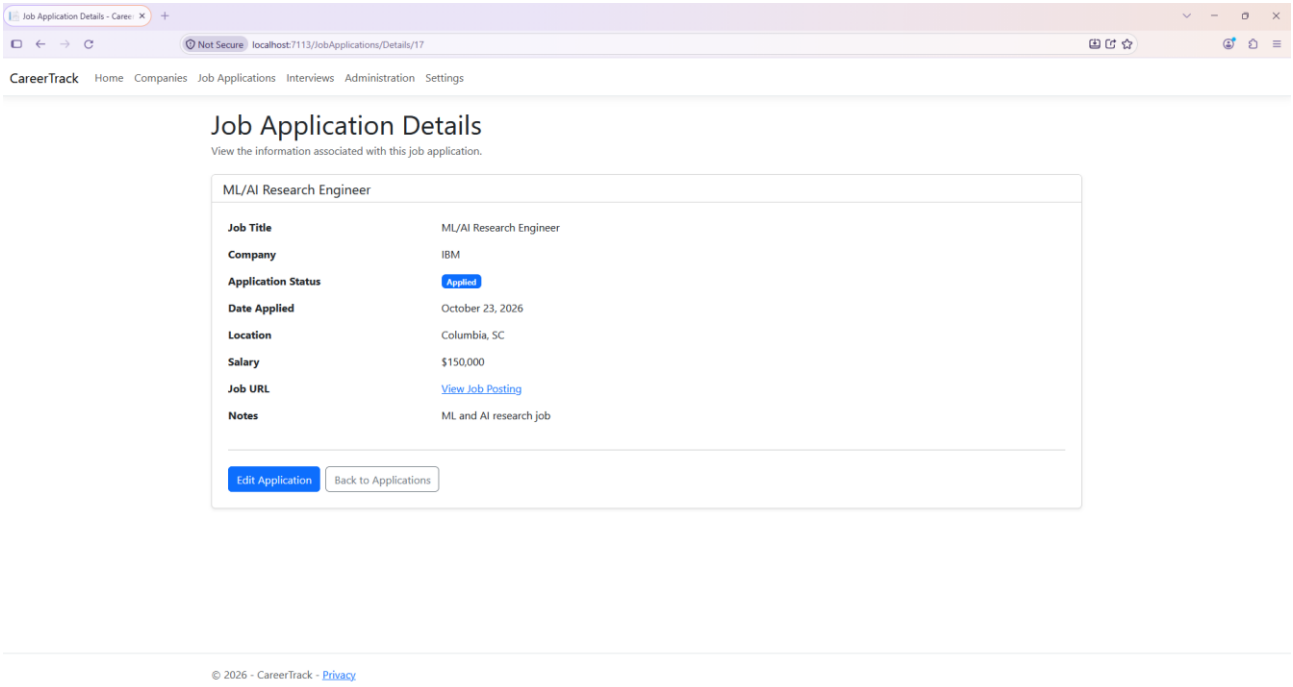


Figure 14. Read Job Application

4.4 Delete

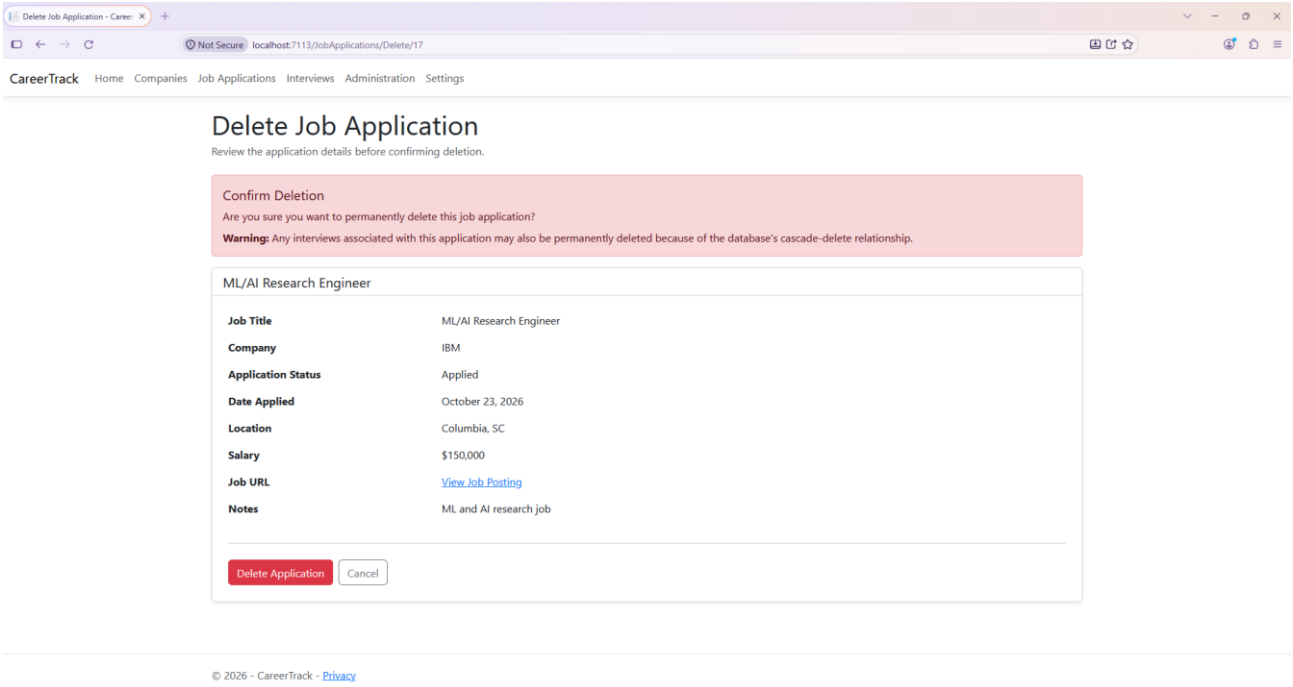


Figure 15. Delete Job Application

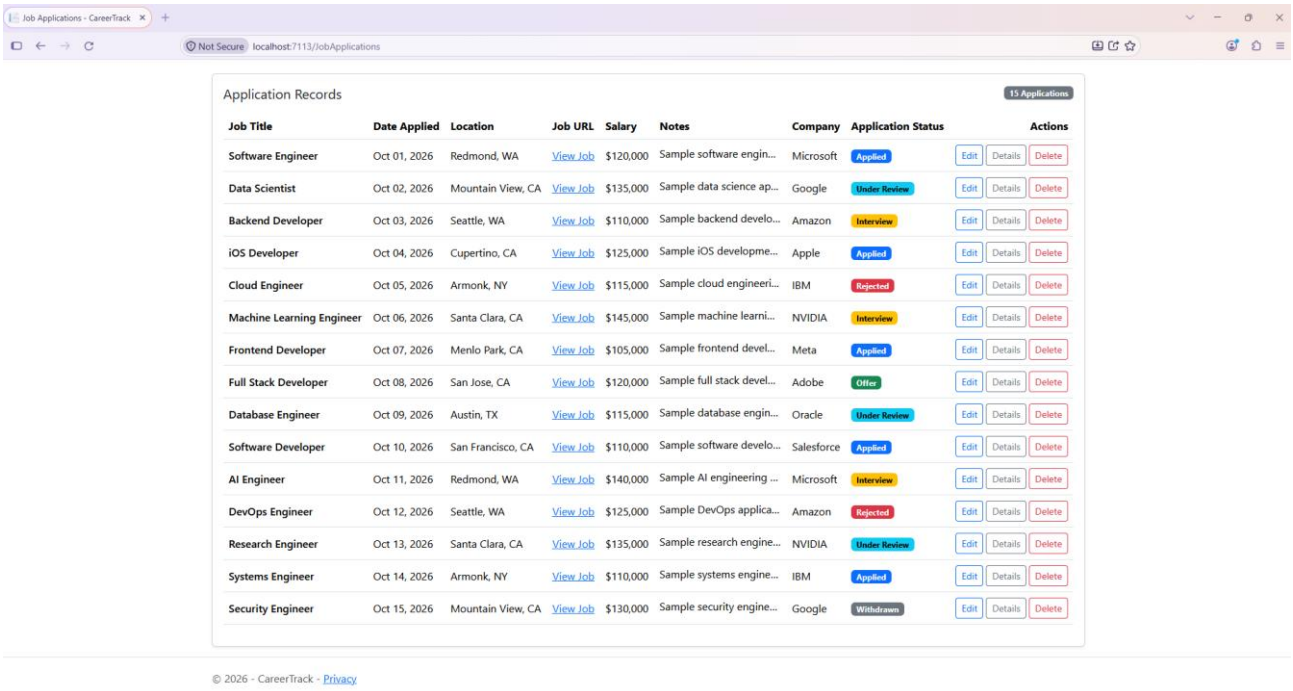


Figure 16. Job Application Deleted

5 — CRUD on Interview Table

5.1 Create

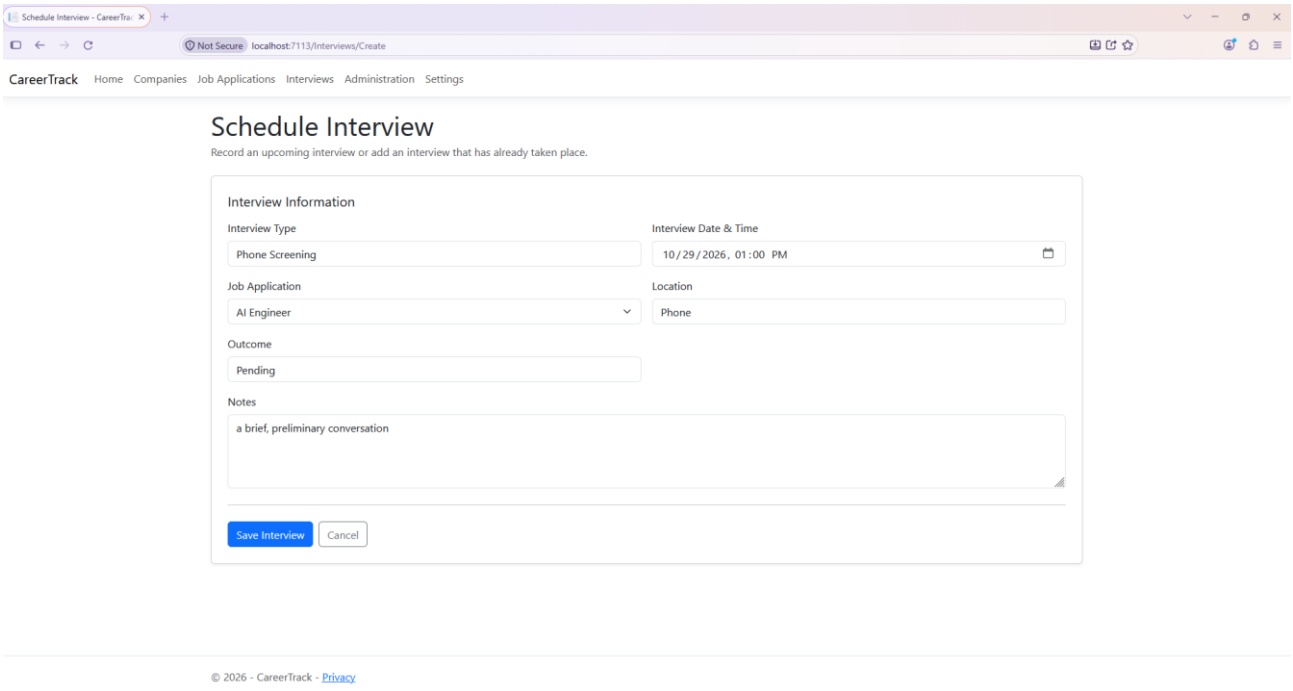


Figure 17. Create Interview

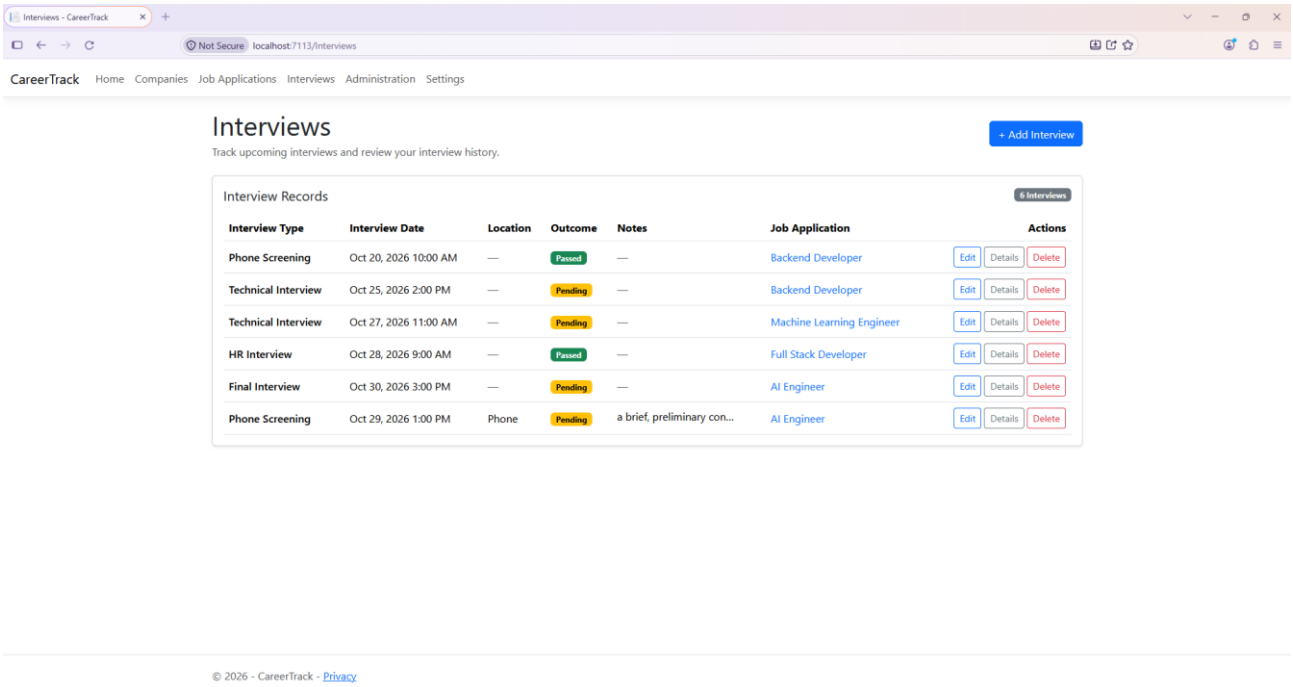


Figure 18. Interview Created

5.2 Update

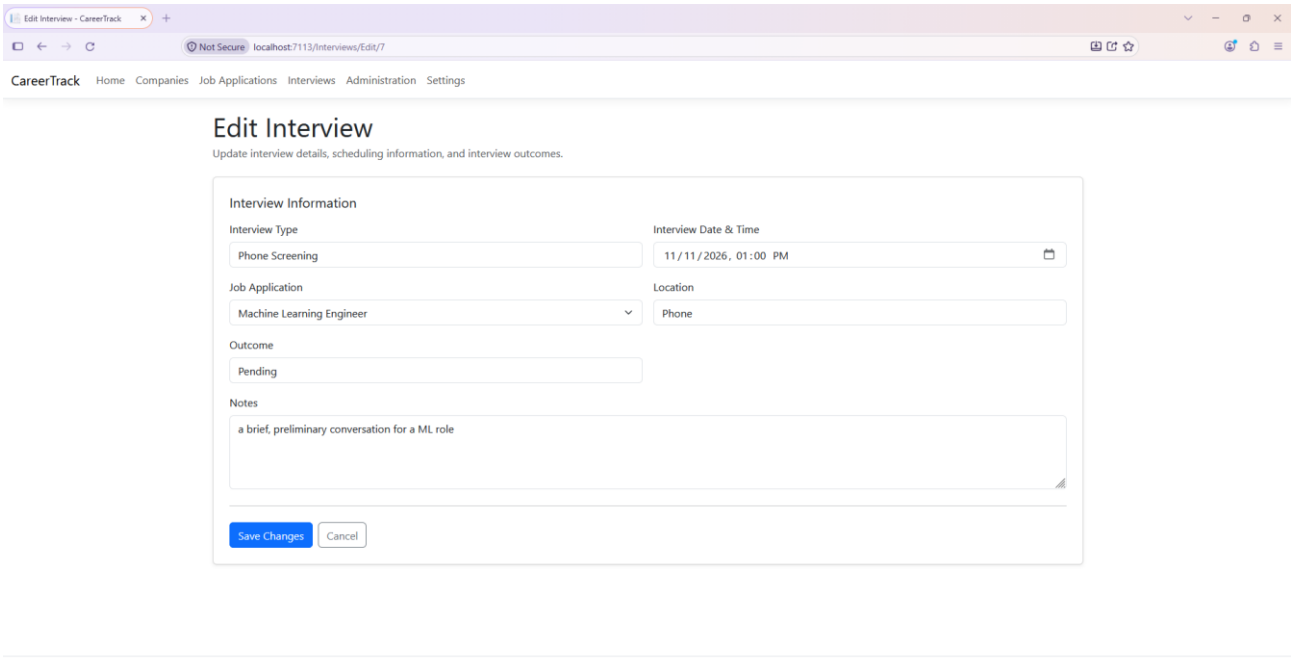


Figure 19. Update Interview

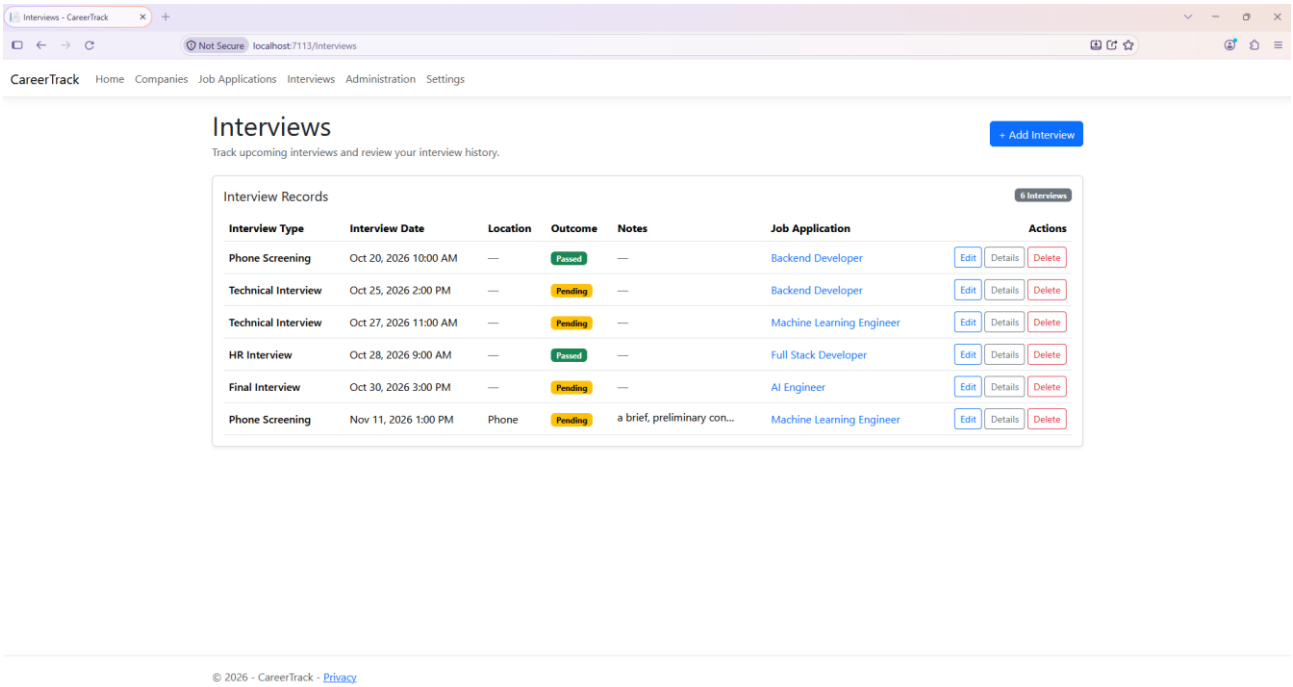


Figure 20. Interview Updated

5.3 Read

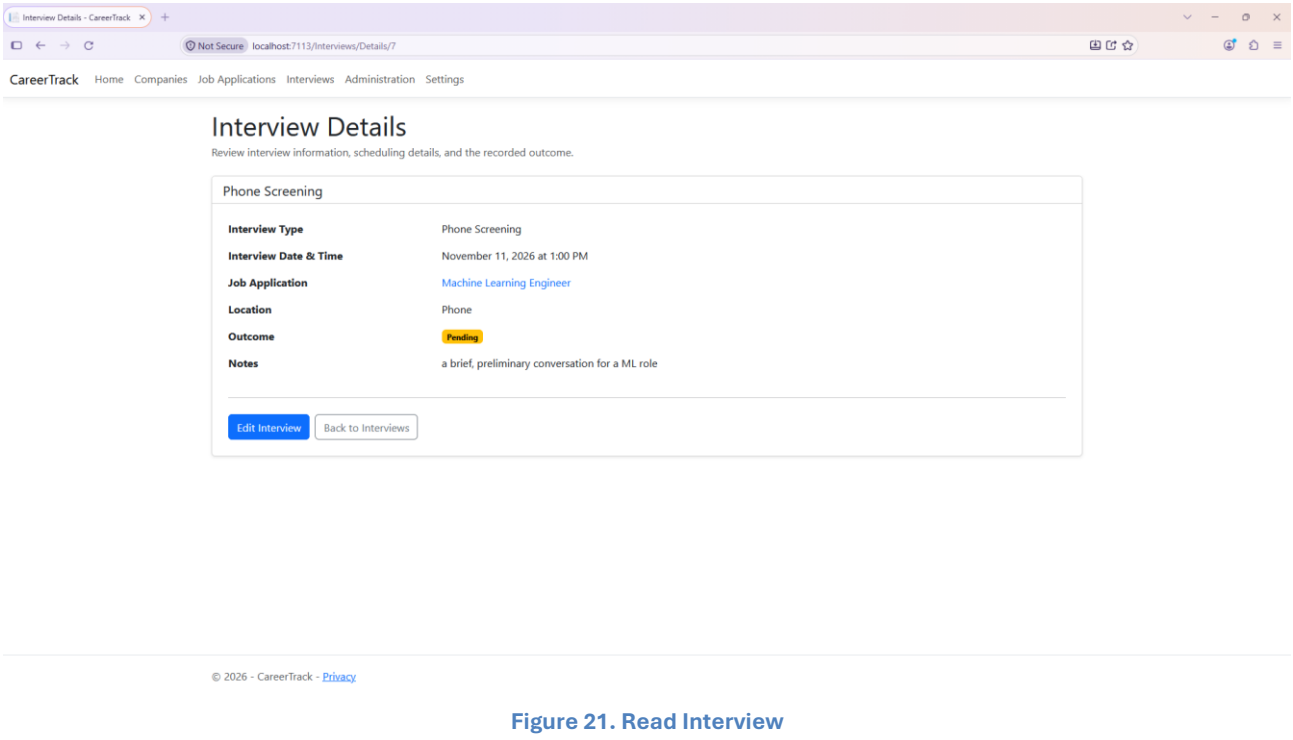


Figure 21. Read Interview

5.4 Delete

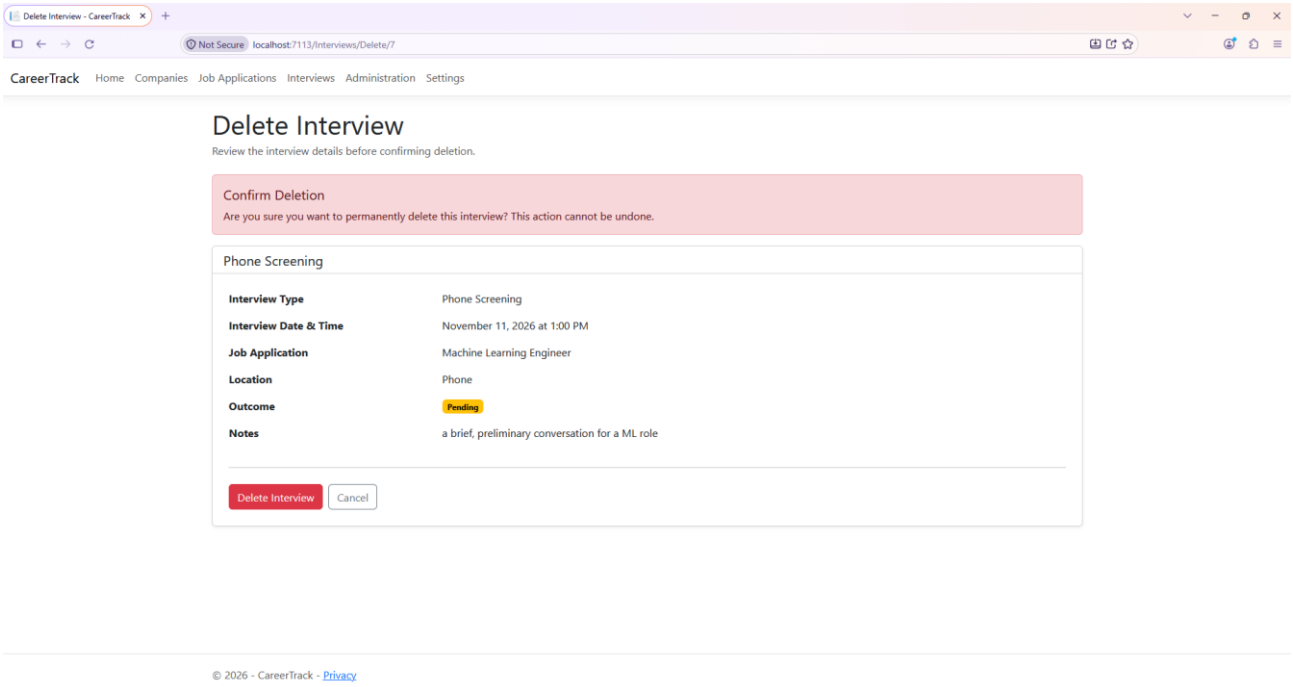


Figure 22. Delete Interview

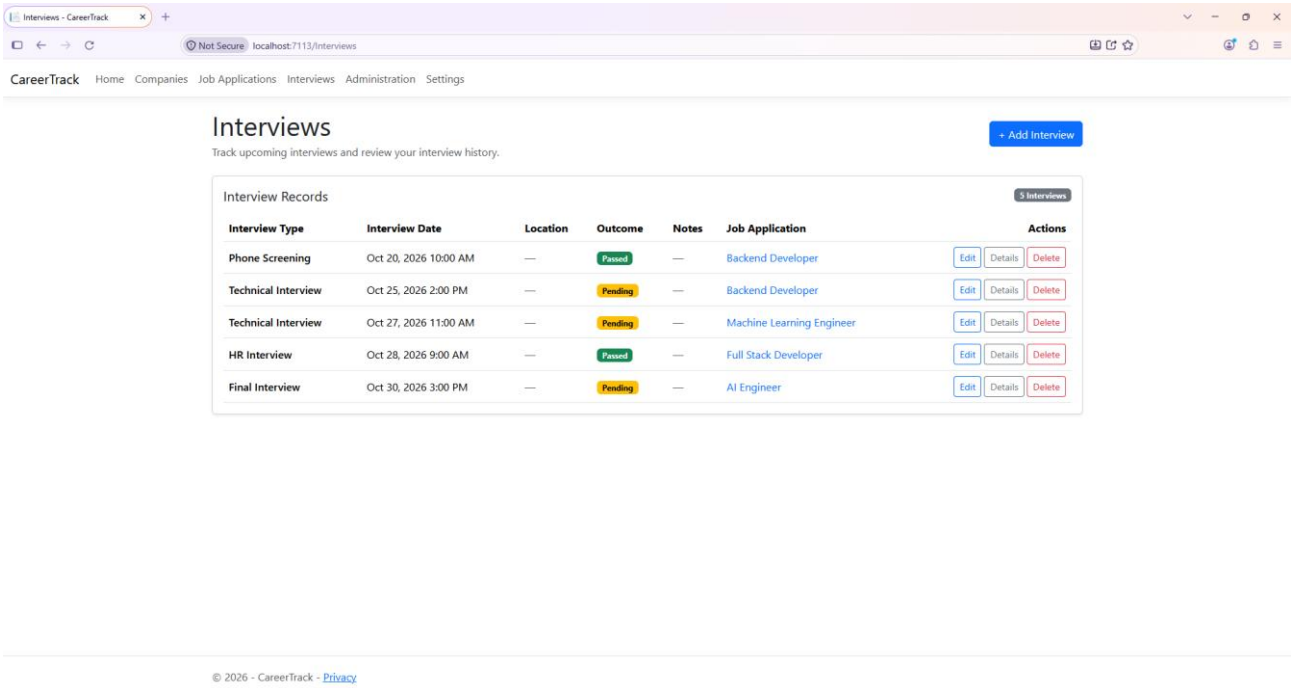


Figure 23. Interview Deleted

6 — Administration

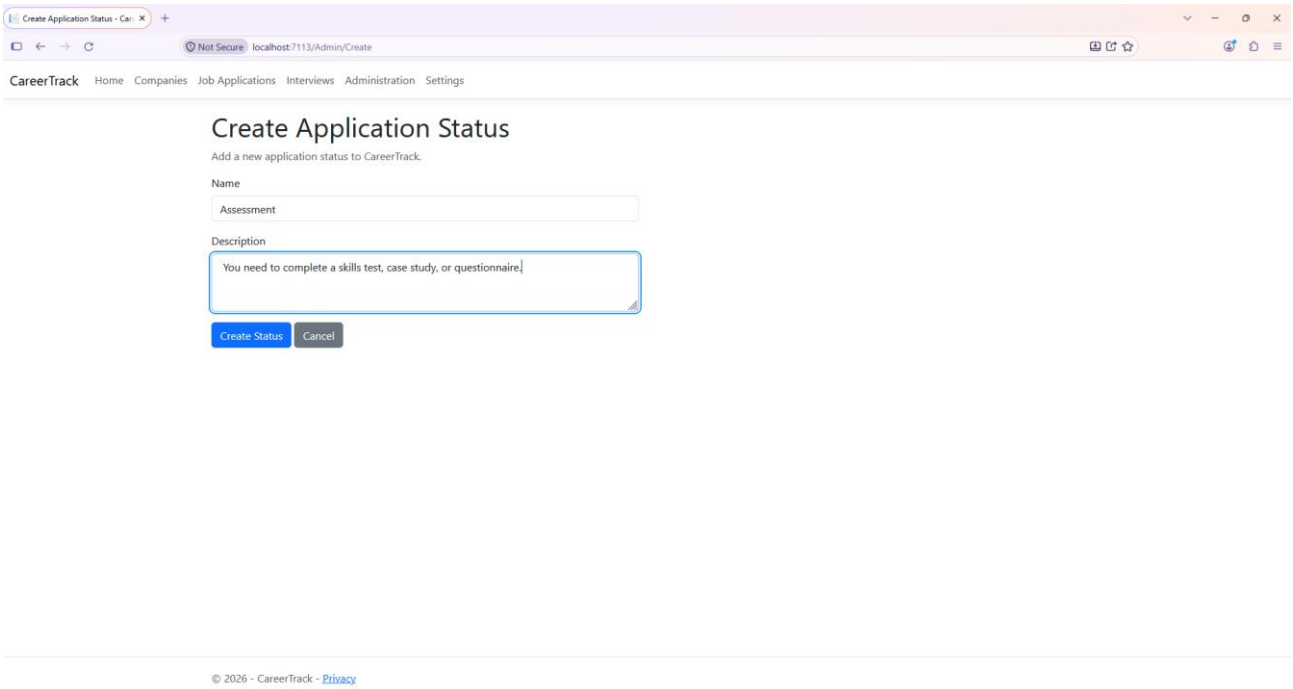


Figure 24. Create Status

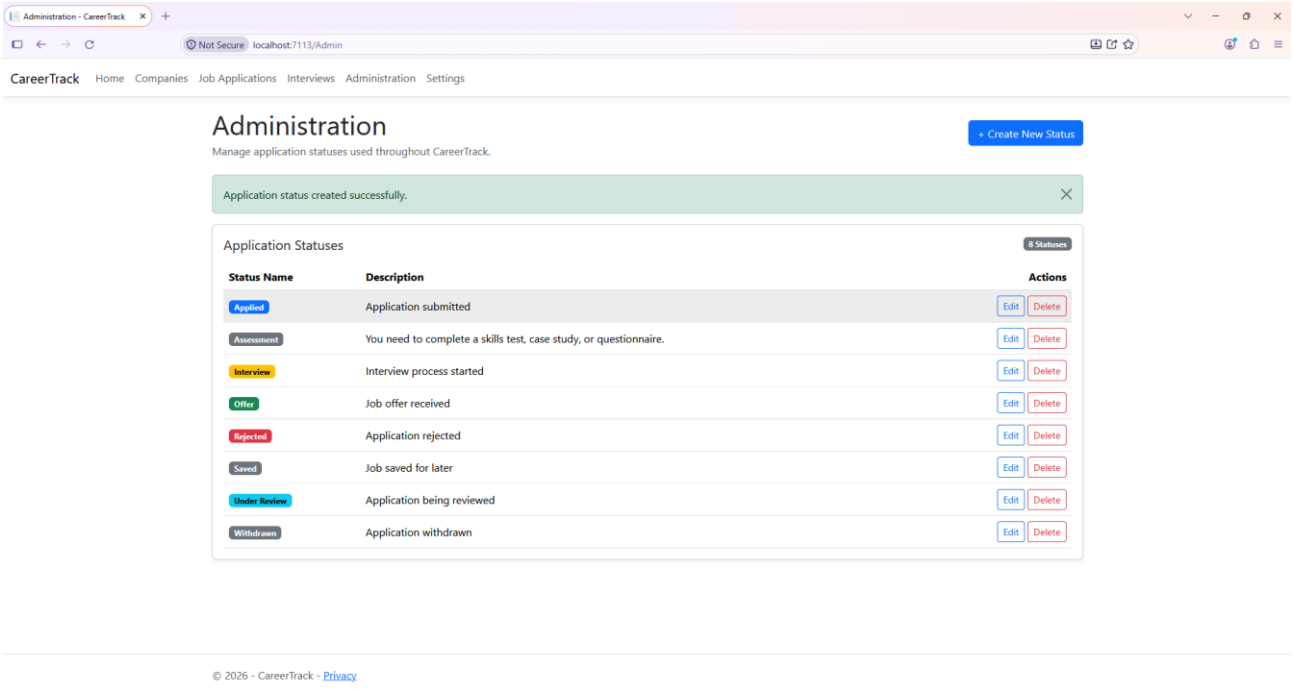


Figure 25. Read Status

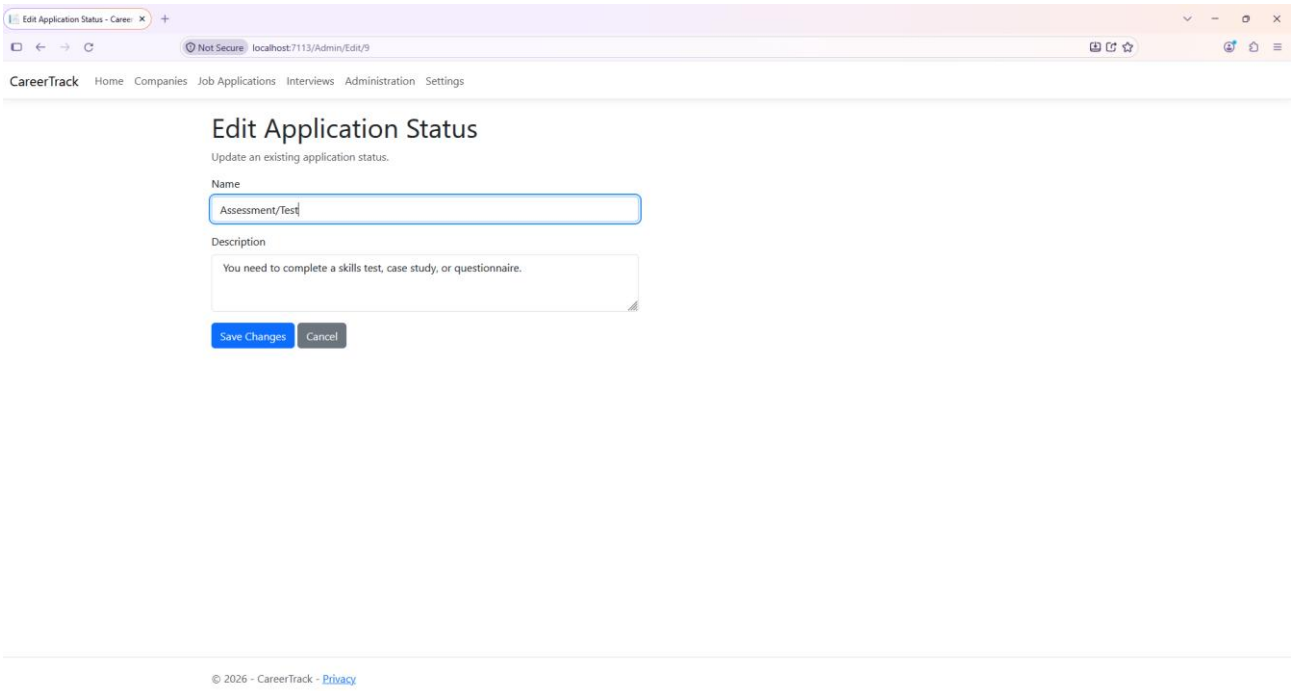


Figure 26. Edit Status

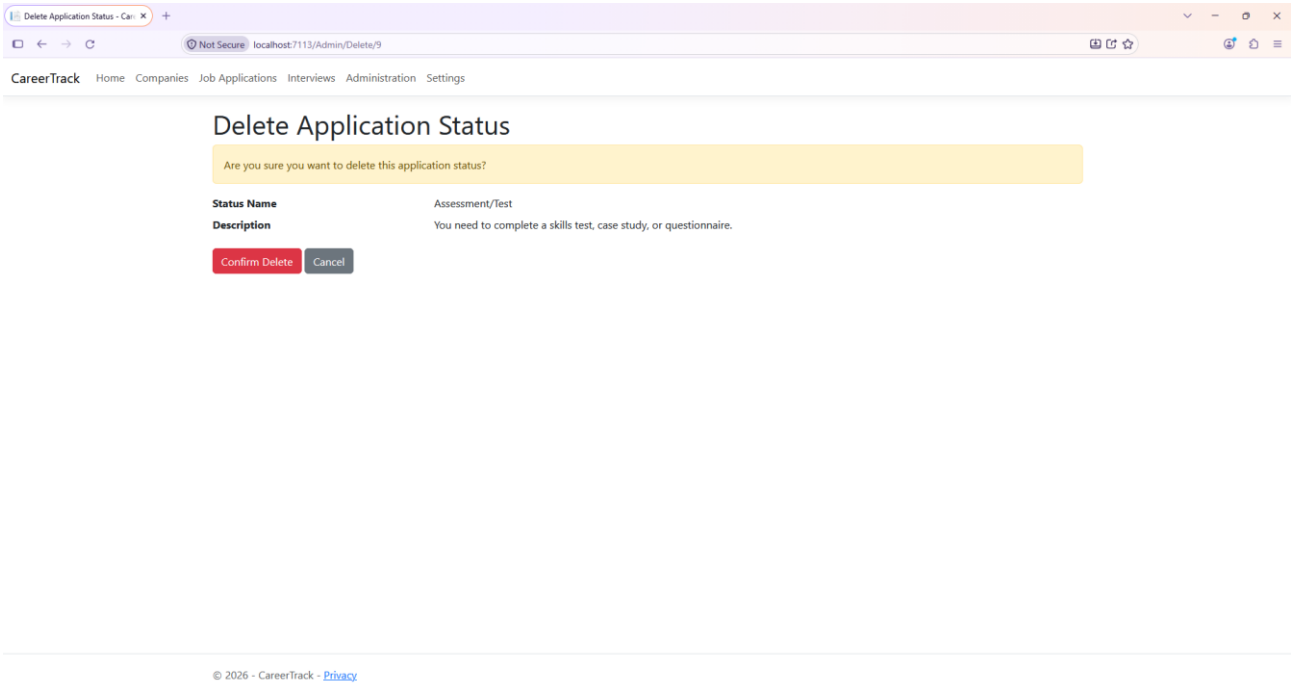
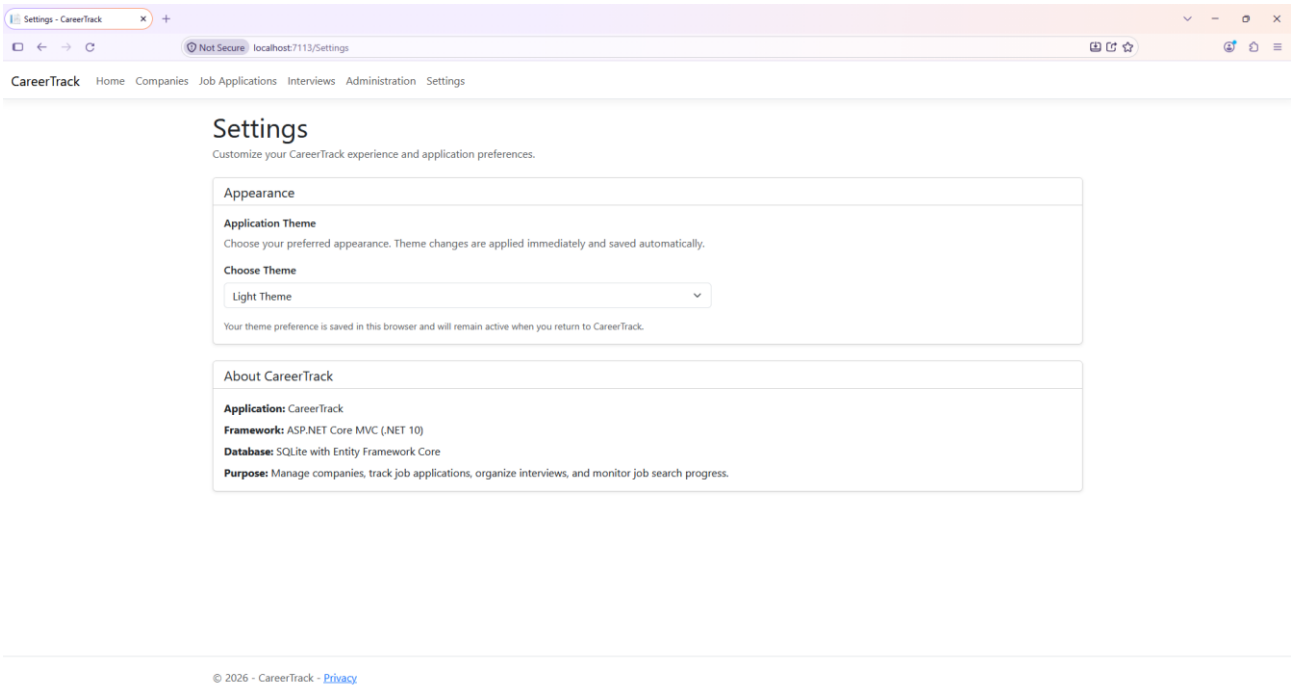
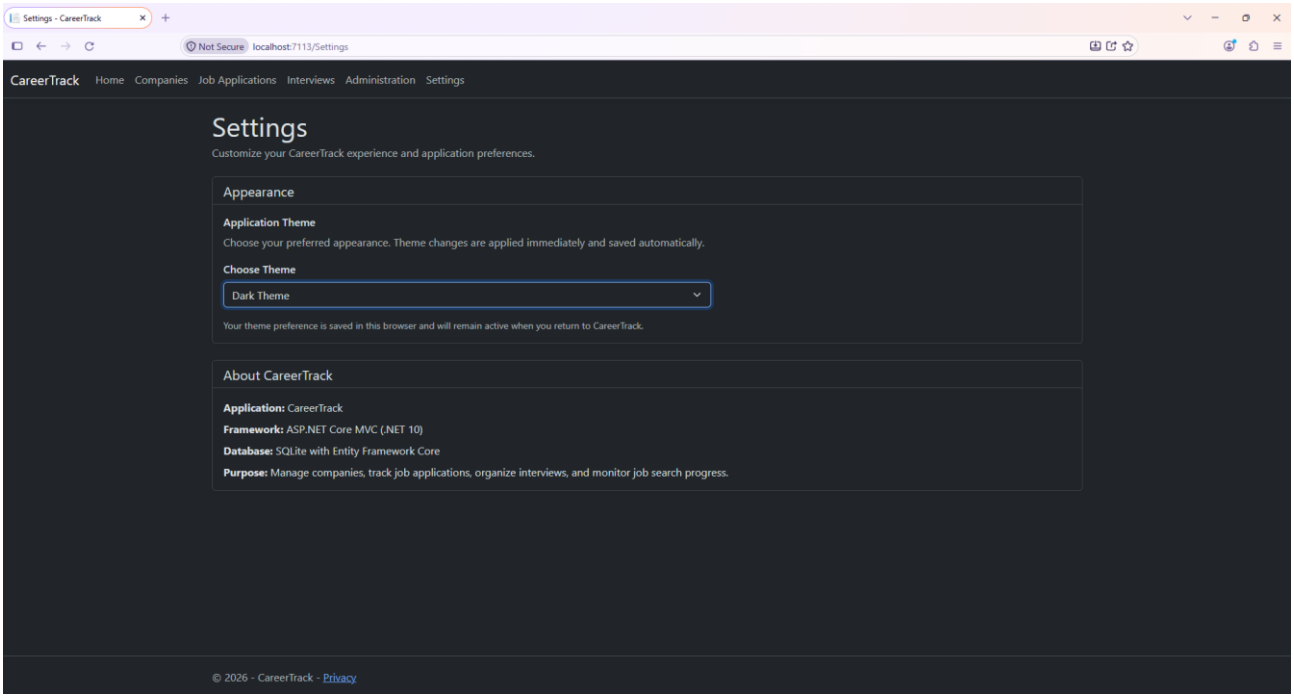
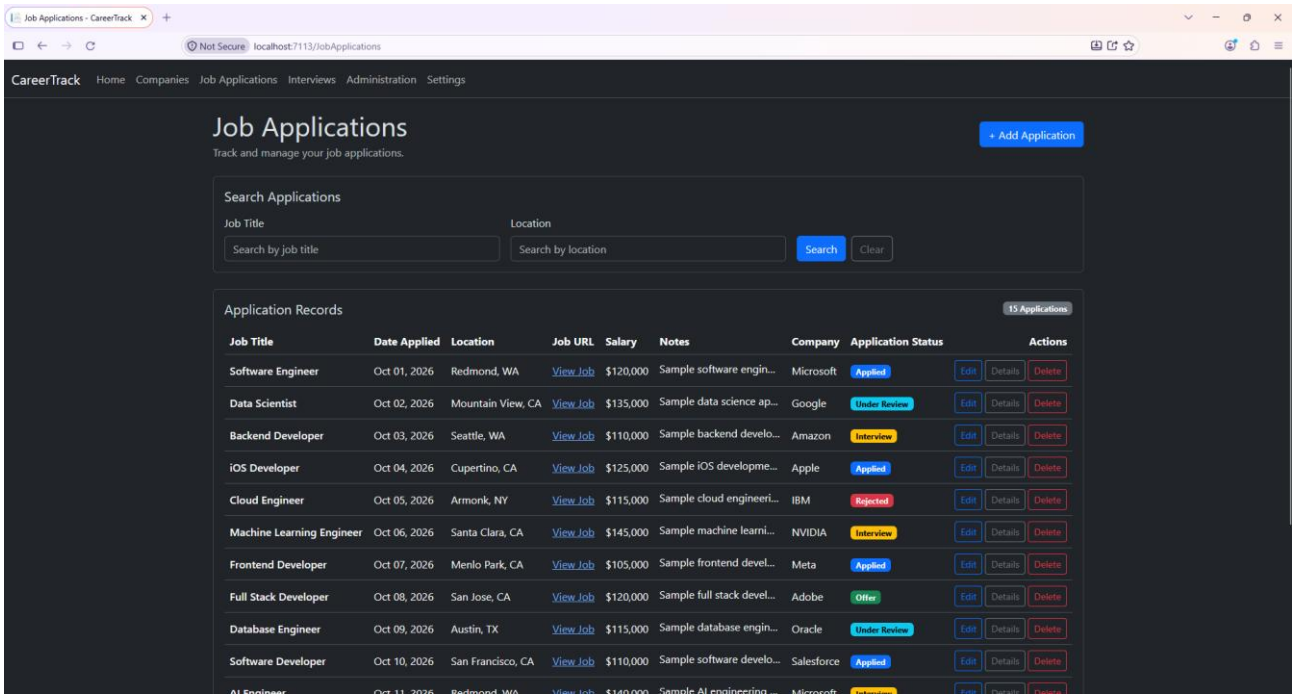


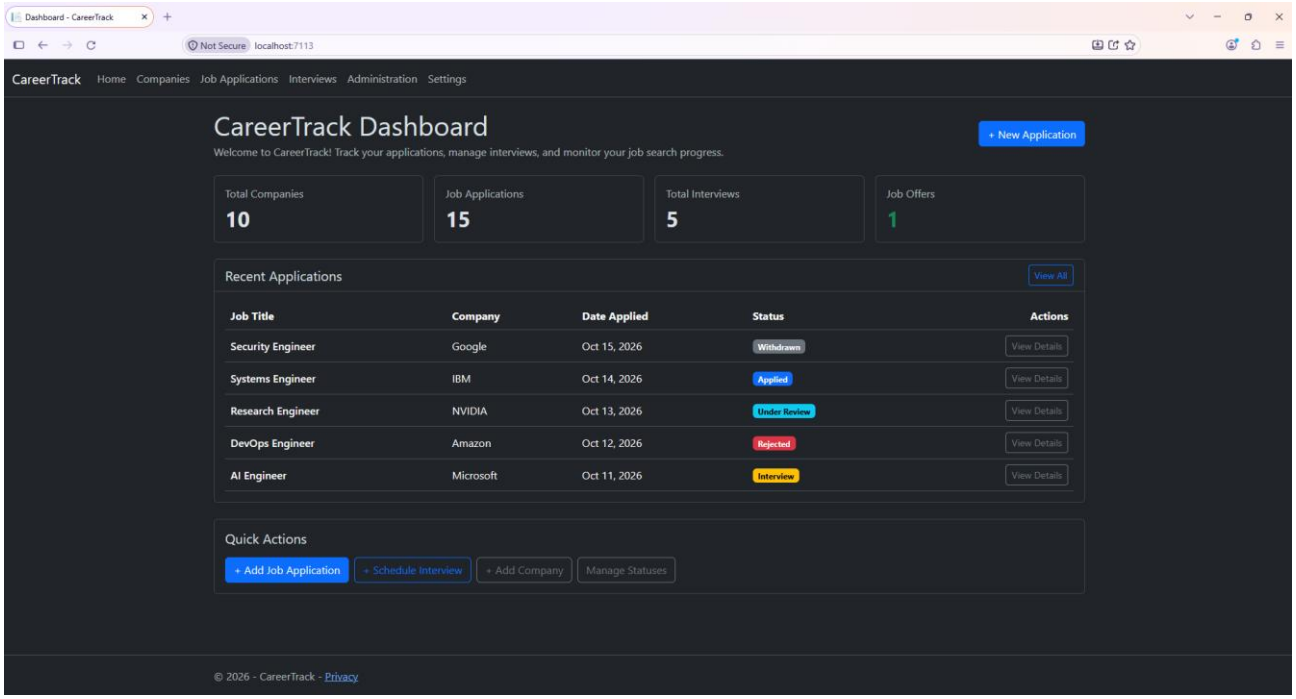
Figure 27. Delete Status

7 — Settings and Theme Switching






Job Applications - CareerTrack
Not Secure localhost:7113/JobApplications
CareerTrack Home Companies Job Applications Interviews Administration Settings
Job Applications
Track and manage your job applications.
+ Add Application
Search Applications
Job Title
Search by job title
Location
Search by location
Search
Clear
Application Records
15 Applications
Job Title | Date Applied | Location | Job URL | Salary | Notes | Company | Application Status | Actions
Software Engineer | Oct 01, 2026 | Redmond, WA | View Job | $120,000 | Sample software engin... | Microsoft | Applied | Edit Details Delete
Data Scientist | Oct 02, 2026 | Mountain View, CA | View Job | $135,000 | Sample data science ap... | Google | Under Review | Edit Details Delete
Backend Developer | Oct 03, 2026 | Seattle, WA | View Job | $110,000 | Sample backend develo... | Amazon | Interview | Edit Details Delete
iOS Developer | Oct 04, 2026 | Cupertino, CA | View Job | $125,000 | Sample iOS developme... | Apple | Applied | Edit Details Delete
Cloud Engineer | Oct 05, 2026 | Armonk, NY | View Job | $115,000 | Sample cloud engineeri... | IBM | Rejected | Edit Details Delete
Machine Learning Engineer | Oct 06, 2026 | Santa Clara, CA | View Job | $145,000 | Sample machine learni... | NVIDIA | Interview | Edit Details Delete
Frontend Developer | Oct 07, 2026 | Menlo Park, CA | View Job | $105,000 | Sample frontend devel... | Meta | Applied | Edit Details Delete
Full Stack Developer | Oct 08, 2026 | San Jose, CA | View Job | $120,000 | Sample full stack devel... | Adobe | Offer | Edit Details Delete
Database Engineer | Oct 09, 2026 | Austin, TX | View Job | $115,000 | Sample database engin... | Oracle | Under Review | Edit Details Delete
Software Developer | Oct 10, 2026 | San Francisco, CA | View Job | $110,000 | Sample software develo... | Salesforce | Applied | Edit Details Delete


Dashboard - CareerTrack
Not Secure localhost:7113
CareerTrack Home Companies Job Applications Interviews Administration Settings
CareerTrack Dashboard
Welcome to CareerTrack! Track your applications, manage interviews, and monitor your job search progress.
+ New Application
Total Companies
10
Job Applications
15
Total Interviews
5
Job Offers
1
Recent Applications
View All
Job Title | Company | Date Applied | Status | Actions
Security Engineer | Google | Oct 15, 2026 | Withdrawn | View Details
Systems Engineer | IBM | Oct 14, 2026 | Applied | View Details
Research Engineer | NVIDIA | Oct 13, 2026 | Under Review | View Details
DevOps Engineer | Amazon | Oct 12, 2026 | Rejected | View Details
AI Engineer | Microsoft | Oct 11, 2026 | Interview | View Details
Quick Actions
+ Add Job Application
+ Schedule Interview
+ Add Company
Manage Statuses
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